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# Severe Drug-Induced Autoimmune Hepatitis Triggered by Cardioril Therapy

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## ABSTRACT

*We report an idiosyncratic case of severe acute drug-induced autoimmune-like hepatitis in a 61-year-old male following 6 weeks of Cardioril (cardioril hydrochloride 40mg daily) administration. Marked hyperbilirubinemia, antinuclear antibody (ANA) seroconversion (titer 1:640), and liver biopsy showing interface hepatitis were documented. Immediate drug withdrawal resulted in clinical and biochemical resolution.*

## INTRODUCTION

Idiosyncratic drug-induced liver injury (DILI) represents a formidable challenge in clinical pharmacology. Cardioril is a novel third-generation antihypertensive agent targeting vascular smooth muscle receptors. While clinical trials reported mild transient aminotransferase elevations in <1.2% of participants, post-marketing surveillance is critical for identifying severe immune-mediated reactions.

## CASE PRESENTATION

A 61-year-old Caucasian male with a 10-year history of refractory hypertension and hyperlipidemia was initiated on Cardioril 40 mg PO QD. Baseline hepatic function panel was completely normal (ALT 22 U/L, AST 19 U/L, Total Bilirubin 0.6 mg/dL). Forty-two days following treatment initiation, the patient presented with progressive fatigue, nausea, dark urine, and jaundice.

Physical examination revealed marked scleral icterus and right upper quadrant tenderness without hepatosplenomegaly or ascites. Laboratory evaluation demonstrated ALT 680 U/L (>12x ULN), AST 510 U/L (>12x ULN), Total Bilirubin 6.2 mg/dL, and Alkaline Phosphatase 240 U/L.

## INVESTIGATIONS & CLINICAL COURSE

Viral serologies for hepatitis A, B, C, and E, cytomegalovirus, and Epstein-Barr virus were non-reactive. Autoimmune serology revealed positive antinuclear antibodies (ANA 1:640, speckled pattern) and elevated IgG levels (2,100 mg/dL). A percutaneous liver biopsy demonstrated severe interface hepatitis with dense portal lymphoplasmacytic infiltrate and bridging necrosis, consistent with drug-induced autoimmune hepatitis.

Cardioril was discontinued on admission. A 4-week tapering course of oral prednisone (40 mg/day initial) was administered. Over the subsequent 6 weeks, serum transaminases and bilirubin normalized completely (ALT 28 U/L, Total Bili 0.8 mg/dL). Re-challenge was not attempted due to severity.

## DISCUSSION & CONCLUSION

This report highlights a probable causal relationship between Cardioril exposure and drug-induced autoimmune hepatitis (RUCAM score = 8, probable). Healthcare practitioners should maintain vigilant liver function monitoring during the first 3 months of Cardioril initiation.

## REFERENCES

1. Chalasani N, et al. Practice Parameters: Evaluation of Drug-Induced Liver Injury. Am J Gastroenterol. 2021;116(5):878-898.
2. Fontana RJ. Pathogenesis of idiosyncratic drug-induced liver injury. Gastroenterology. 2022;162(5):1370-1388.